

Exam Date & Time: 31-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VII Semester - Make up Examination - December -2024

Human Interface Interaction [IT-4049]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze how Fitts' Law can be applied to enhance user interface design by optimizing the placement and size of interactive elements. Evaluate how changes in element size and distance affect user speed and accuracy, and provide examples that demonstrate how Fitts' Law can improve usability and user experience. (5)
 - A)
 - B) In the context of an online banking system, demonstrate how the relationship between the domain (banking), the user's goal (e.g., transferring money), and the tasks (steps involved in completing the transaction) shapes the design of the system's interface and functionality. (3)
 - C) Examine the differences between self perception and person perception. (2)
- 2) Scenario: You are tasked with designing a new feature for an online learning platform that allows instructors to create and share interactive quizzes. The goal is to enhance the learning experience and increase engagement through interactive content. Constraints include a limited budget, the need to integrate with existing learning management systems, and a requirement to support users with varying levels of internet connectivity. There are trade-offs between adding complex features, such as real-time feedback for quiz takers, and ensuring the platform remains accessible to users with slower connections. (5)

Evaluate how you would address the goals, constraints, and trade-offs involved in designing the interactive quiz feature for the online learning platform. How would you balance engagement with accessibility and integration requirements while managing resource limitations? Discuss how these factors would shape the design of the new feature.

 - B) Demonstrate with a brief example the steps involved in the process of design. (3)
 - C) Utilize two principles of universal design and explain how they influence the creation of effective user interfaces. Provide example of how each principle can be applied to improve usability and user experience. (2)
- 3) A company wants to create a coding and learning platform, LeCode for Engineering pursuing students. Identify whether altering the layout and color scheme of its website improves user readability and time spent on the site. (5)
 - A)

- B) Illustrate the advantages of using Participatory Design. (3)
- C) Categorize the rating scale used by evaluators in UX design evaluations. (2)
- 4) Evaluate how the Prescriptive aspect of GOMS can guide the development of a training strategy for the given scenario. Propose an effective strategy to improve task efficiency by encouraging the use of the faster method.
- A) Scenario:
A hospital's system offers two logout methods: (5)
1. Method 1: Click the "Logout" button.
 2. Method 2: Use the Ctrl + L shortcut.
- A GOMS analysis reveals that new employees prefer the first method, while experienced employees prefer the second.
- B) Compare between competence models and performance models in interaction design. (3)
- C) According to Nielsen's experience, the typical number of evaluators sufficient for identifying usability problems and what percentage of overall usability issues can they identify. (2)
- 5) Evaluate the impact of augmented reality on live musical performances by examining the use of virtual effects that enhance audience experiences.
- i. Illustrate how AR can create an immersive atmosphere during the performance.
 - ii. Assess the potential effects of this technology on audience engagement and satisfaction. (5)
- Identify one drawback of integrating AR into live entertainment
- B) Explain in brief what you understand by PAC architectural pattern (3)
- C) Deduce the techniques used in Observational Methods of Evaluation through user participation. (2)

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